Research Statement

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We live in a world where agents do not act in isolation. Solving problems in complex environments requires agents to coordinate effectively, understand what others can do, and adapt their behaviour accordingly. I am interested in **identifying and modelling the core computational principles underlying goal-directed collaboration**, and ultimately in **building collaborative AI systems based on these principles**. To that end, just as we built aeroplanes by understanding the principles of aerodynamics rather than imitating wing-flapping, I aim to build collaborative AI systems by understanding principles from biological collaboration rather than directly imitating biological mechanisms.

In particular, biological systems demonstrate these principles across species and contexts. From simple organisms to humans, agents have evolved sophisticated methods for collaboration, enabling them to achieve shared goals through mutual understanding and coordinated actions. Consider mixed-species bird flocks: warblers follow chickadees to locate food, while chickadees benefit from increased group vigilance. Similarly, in human collaboration such as pair programming, one programmer focuses on implementation while the other maintains a broader perspective.

Artificial agents, however, lack the collaborative abilities needed for flexible, goal-directed coordination. My research vision is to address this gap by developing agents with causal world models that support reasoning about how other agents' mental states drive behaviour, mechanisms for active information gathering to reduce uncertainty about collaborators' goals and abilities, and principled methods for efficient belief updating as new evidence emerges during interaction.

In what follows, I outline my past contributions towards this vision and future directions for building AI systems capable of genuine collaboration.

Past Experience and Contributions

My time as a research trainee at Basis Research Institute provided me with the technical foundations and research experience essential for pursuing my research vision. Working closely with Zenna Tavares, who advised the projects outlined below, I developed skills in probabilistic reasoning, causal inference, and empirical evaluation.

Collaborative Intelligent Systems During my time at Basis, I regularly participated in weekly meetings led by Emily Mackevicius, a neuroscientist working on multi-agent behaviour and communication. I helped prototype models that infer agents' goals from observed behavior in simple organisms, like slime mold. Through these meetings and informal conversations, I developed my interest in multi-agent collaboration and communication, which became central to my research vision.

Probabilistic Programming and Causal Inference Probabilistic modelling lies at the foundation of understanding these agent behaviours. To solidify my skills and knowledge, I worked on advancing the probabilistic programming infrastructure at Basis. Specifically, I developed *CounterfactualFairness.jl*, a Julia package implementing counterfactual reasoning methods for algorithmic fairness. I translated ProbMods [2] from WebPPL to Omega [5], fixing issues in Omega to ensure the examples worked correctly. I also implemented *parametric inversion* [4], a program transformation technique that inverts non-injective functions by introducing parameters, in JAX, enabling automatic inversion of simulation models like Lotka-Volterra and boids.

Resource-Rational Abstraction As my first attempt at a project involving modelling agents, I led a reading group on abstraction, where we read papers on causal abstraction theory [1] and cognitive science research on how humans form abstract mental models based on the queries they are asked [3]. From these discussions, I developed a framework for automatically constructing model abstractions that balance computational cost with the accuracy needed for specific queries. I formalised this as an optimisation problem and implemented it in MuJoCo, a physics engine, where geometric abstractions were automatically selected based on query requirements. This work taught me to formalise cognitive principles mathematically and implement them as concrete algorithms. I want to apply a similar approach to collaborative AI systems: identify principles from biological collaboration, formalise them mathematically, and implement them as practical algorithms.

Evaluating World-Model Learning [6] Finally, during my research and reading, I realised the importance of world models within an agent. However, I identified a gap in the existing literature: there was no consensus on how to evaluate world modelling. Therefore, I started work on AutumnBench, a concrete benchmark with 43 grid-world environments and 129 tasks, designed to measure how agents learn world models through reward-free exploration and testing on derived challenge tasks. I also developed WorldTest, a theoretical framework that formalises the evaluation approach underlying AutumnBench. I collected data comparing human and LLM performance on AutumnBench through statistical analyses. This work, currently under review at ICLR, provided me with experience in designing empirical studies, formalizing research questions into concrete benchmarks, and working with experimental data. Understanding how agents learn world models is foundational to my research goals: modelling other agents as intentional systems requires constructing structured representations of both the environment and the causal relationships within it.

These projects collectively equipped me with technical skills in probabilistic and causal reasoning, experience in translating cognitive principles into computational frameworks, and knowledge in designing and conducting empirical studies—abilities that will enable me to pursue my vision of building collaborative AI systems grounded in principled computational approaches.

Future Research Directions

To understand and build collaborative AI systems, I am interested in exploring how agents can construct rich models of other agents and their environment, efficiently update beliefs through interaction, and learn from the collaborative principles exhibited by biological systems. I see several interconnected challenges that I would be excited to pursue:

Building explicit models of other agents. Current AI systems lack structured representations of other agents as intentional systems with distinct goals, beliefs, and abilities. I would like to investigate how agents can construct compositional world models that explicitly represent other agents' latent states and potential behaviours, and develop frameworks where agents maintain structured beliefs over other agents' policies. A key challenge I would like to address is learning efficient abstractions of other agents' behaviour and determining when to model another agent as pursuing a goal versus following a simple heuristic.

Efficient belief updating and information gathering. Effective collaboration requires not only building models of other agents, but also updating the models efficiently as new evidence emerges and actively gathering information to reduce uncertainty. My work on AutumnBench reveals that LLMs currently struggle to update beliefs when faced with contradictory evidence—they persist with initial hypotheses even when observations clearly violate the learned rules. I am interested in developing meta-reasoning frameworks that enable agents to quantify uncertainty about their models and use them to update their beliefs.

Understanding collaborative cognition in biological agents. To build AI systems that can collaborate effectively, we must first understand the computational principles underlying collaborative intelligence in biological systems. I would like to design and conduct behavioural experiments that reveal how biological agents—from humans to birds to other animals—build models of collaborators during interaction.

I am excited by each of these research directions and would welcome the opportunity to explore them, either individually or in combination. The interdisciplinary nature of ELLIS would provide an ideal environment to pursue these questions and discover connections I haven't yet envisioned.

References

[1] Sander Beckers and Joseph Y. Halpern. Abstracting causal models. In *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence*, 2019.

[2] Noah D Goodman, Joshua B. Tenenbaum, and The ProbMods Contributors. Probabilistic Models of Cognition. http://probmods.org/v2, 2016.

[3] Steven M Shin and Tobias Gerstenberg. Learning what matters: Causal abstraction in human inference. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2023.

[4] Zenna Tavares and Armando Solar-Lezama. Parametric inverse simulation. In *Advances in Approximate Bayesian Inference*, 2016.

[5] Zenna Tavares, James Koppel, Xin Zhang, Ria Das, and Armando Solar-Lezama. A language for counterfactual generative models. In *International Conference on Machine Learning*. Proceedings of Machine Learning Research, 2021.

[6] Archana Warrier, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B. Tenenbaum, Sebastian Vollmer, Kevin Ellis, and Zenna Tavares. Benchmarking world-model learning via environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.